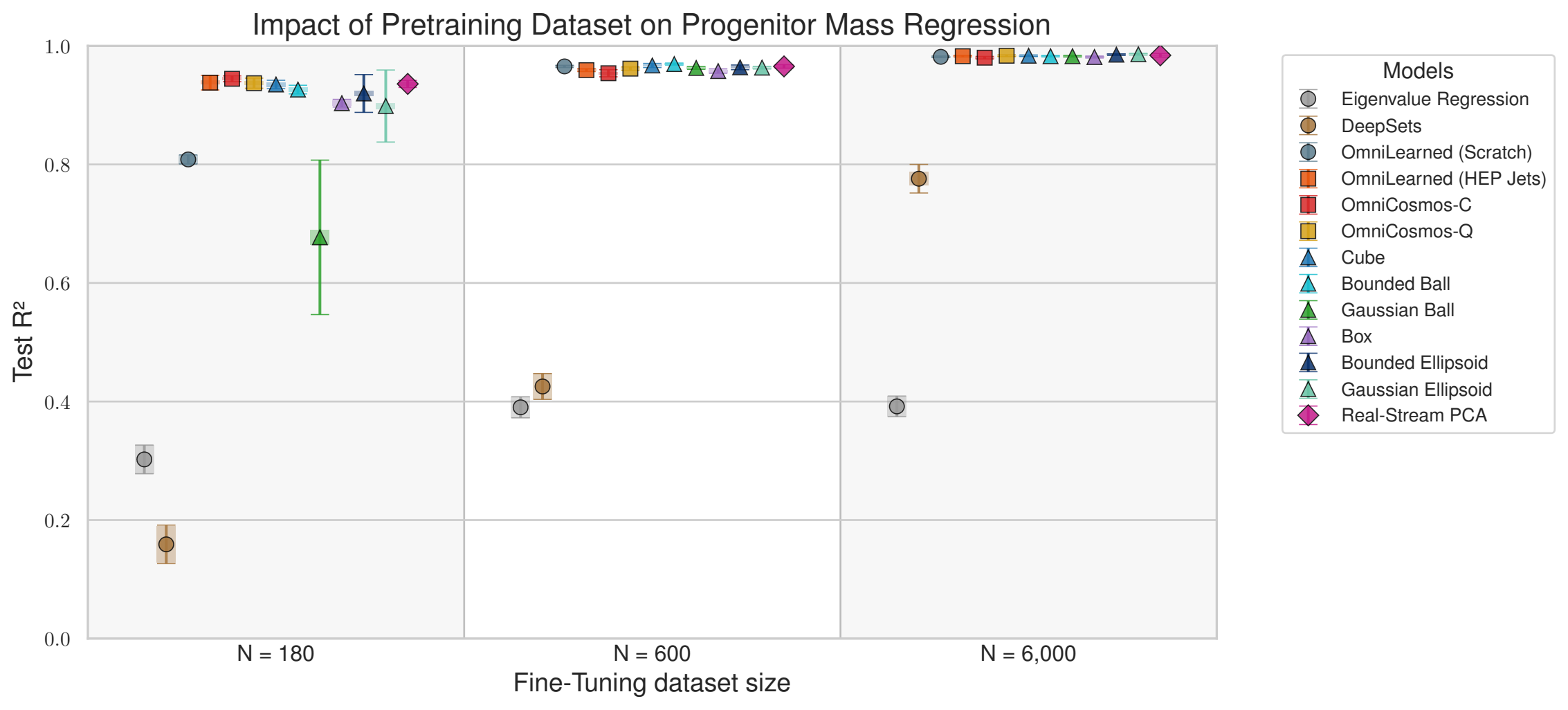

Impact of Pretraining Dataset on Progenitor Mass Regression
Test R²
1.0
0.8
0.6
0.4
0.2
0.0
N = 180
N = 600
N = 6,000
Fine-Tuning dataset size
Models
Eigenvalue Regression
DeepSets
OmniLearned (Scratch)
OmniLearned (HEP Jets)
OmniCosmos-C
OmniCosmos-Q
Cube
Bounded Ball
Gaussian Ball
Box
Bounded Ellipsoid
Gaussian Ellipsoid
Real-Stream PCA